



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to Dhanalakshmi Agencies Adhoc

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

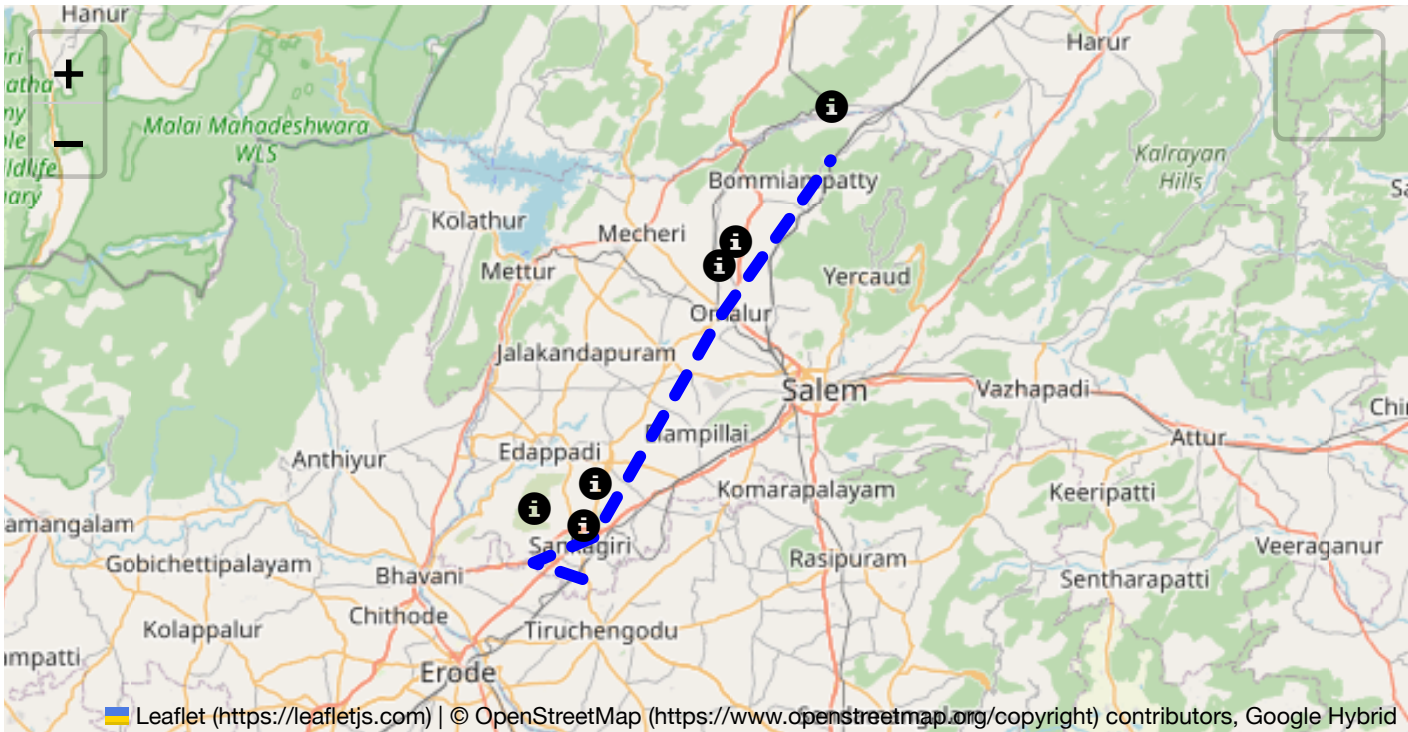
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 87.82 km
Estimated Duration: 2.1 hours

Adjusted Duration (Heavy Vehicle): 2.6 hours

Start: (11.4381, 77.8734)

End: (11.919589, 78.165205)



Welcome to the Journey Risk Management Study

Certainly, here is a detailed safety analysis of the route from CVQF+23W, Sangagiri to W5C8+467, Lokur, Tamil Nadu, focusing on the key concerns for driving heavy vehicles carrying hazardous materials.

1. Overview of the Route Map:

- The route covers approximately 87.82 kilometers, beginning in Sangagiri and passing through Bhavani Main Road, Padaiveedu, SH 86 in Aiveli, and Salem before reaching the destination in Lokur. Key roadways include NH44, SH86, and smaller connecting roads.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

- The region experiences a tropical climate with high temperatures most of the year. Monsoon season is from June to September, which can lead to heavy rainfall and potential flooding, especially in the low-lying areas. This may result in slippery roads and reduced visibility.

3. Analysis of Traffic Patterns:

- Traffic tends to be heavier during morning and evening peak hours (8-10 AM and 5-8 PM) around Salem. Bhavani Main Road and NH44 in Salem are known for congestion due to heavy local and commercial traffic.

4. Assessment of Road Quality and Infrastructure:

- Major highways like NH44 and SH86 are generally well-maintained, but some patches, especially on rural roads, may have potholes or uneven surfaces. Regular maintenance updates should be checked before trips.

5. Suggestions for Alternative Routes for Emergencies:

- In case of a blockage on the main route, an alternative could be taking state highways branching from SH86 or NH544 between locations. Use of local routes should be verified for road quality.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Vehicles transporting hazardous materials need proper documentation and adherence to specific route restrictions in populated and environmentally sensitive areas. Night travel might be restricted in certain zones.

7. Overview of Historical Incidents:

- Historical incidents on this route have mostly been related to vehicular collisions. Limited data on incidents specific to hazardous materials suggest the need for cautious transit through Salem due to high traffic.

8. Environmental Considerations and Sensitive Areas:

- The route passes through agricultural zones. It is crucial to avoid spills and emissions to protect these areas. Specific zones near water bodies must be protected from contamination.

9. Analysis of Communication Coverage:

- Most areas along NH44 have good mobile coverage, but some rural stretches between towns may experience weak signals. It is important to have reliable communication equipment onboard.

10. Estimated Emergency Response Times:

- Emergency services in major towns like Salem are equipped for quick response, estimated at 15-30 minutes. In rural or less accessible stretches, response times may reach up to 45 minutes to an hour.

11. Overall Summary of Risk Assessment:

- The route generally poses moderate risk, with congestion in Salem and potential weather impacts being major considerations. While infrastructure supports heavy vehicle travel, drivers should be cautious of rural road conditions and ensure compliance with regulations. Preparation for communication challenges and swift action in emergencies is advised to ensure safety for hazardous material transport.

Engaging with local transport authorities for current updates and potential roadwork information is essential before commencing the journey. Regularly maintained vehicles and well-trained drivers will significantly mitigate risks along this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	Medium	11.44900, 77.87406	30 KM/Hr
3	Turn	High	11.45353, 77.85697	15 KM/Hr
4	Turn	Medium	11.45853, 77.85139	30 KM/Hr
5	Turn	Medium	11.45906, 77.85132	30 KM/Hr
6	Turn	High	11.46307, 77.84941	15 KM/Hr
7	Turn	High	11.45542, 77.81725	15 KM/Hr
8	Turn	High	11.45636, 77.81678	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
9	Turn	High	11.49224, 77.89606	15 KM/Hr
10	Turn	High	11.49153, 77.89615	15 KM/Hr
11	Blind Spot	Blind Spot	11.48364, 77.88780	10 KM/Hr
12	Turn	Medium	11.49155, 77.88776	30 KM/Hr
13	Turn	Medium	11.49441, 77.88495	30 KM/Hr
14	Turn	Medium	11.49469, 77.88488	30 KM/Hr
15	Turn	High	11.55614, 77.89764	15 KM/Hr
16	Turn	Medium	11.58218, 77.90845	30 KM/Hr
17	Turn	Medium	11.58285, 77.90790	30 KM/Hr
18	Turn	High	11.58330, 77.90642	15 KM/Hr
19	Turn	Medium	11.64217, 77.95200	30 KM/Hr
20	Turn	Medium	11.64712, 77.95118	30 KM/Hr
21	Turn	Medium	11.67858, 77.96834	30 KM/Hr
22	Turn	High	11.70218, 77.97866	15 KM/Hr
23	Turn	Medium	11.71917, 78.01123	30 KM/Hr
24	Turn	Medium	11.72565, 78.01770	30 KM/Hr
25	Turn	Medium	11.73787, 78.03528	30 KM/Hr
26	Turn	High	11.73651, 78.03868	15 KM/Hr
27	Turn	Medium	11.74120, 78.04588	30 KM/Hr
28	Turn	Medium	11.78045, 78.05792	30 KM/Hr
29	Blind Spot	Blind Spot	11.87104, 78.08356	10 KM/Hr
30	Turn	Medium	11.87106, 78.08579	30 KM/Hr
31	Turn	Medium	11.87257, 78.08738	30 KM/Hr
32	Turn	Medium	11.86902, 78.12044	30 KM/Hr
33	Turn	Medium	11.86857, 78.12144	30 KM/Hr
34	Turn	Medium	11.86477, 78.12563	30 KM/Hr
35	Turn	High	11.85820, 78.13580	15 KM/Hr
36	Turn	Medium	11.86060, 78.13954	30 KM/Hr
37	Turn	Medium	11.86380, 78.14016	30 KM/Hr
38	Turn	Medium	11.86659, 78.13878	30 KM/Hr
39	Turn	Medium	11.88120, 78.14383	30 KM/Hr
40	Turn	Medium	11.88400, 78.14472	30 KM/Hr
41	Turn	Medium	11.88515, 78.14462	30 KM/Hr
42	Turn	Medium	11.88745, 78.14560	30 KM/Hr
43	Turn	Medium	11.90585, 78.15291	30 KM/Hr
44	Turn	Medium	11.90748, 78.15209	30 KM/Hr
45	Turn	Medium	11.90783, 78.15208	30 KM/Hr
46	Turn	Medium	11.91206, 78.15580	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
47	Turn	Medium	11.91299, 78.15608	30 KM/Hr
48	Turn	Medium	11.91510, 78.15822	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
6	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
7	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
8	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
9	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
13	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
14	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
15	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
16	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
17	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
18	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
19	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
20	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
21	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
25	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
27	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
28	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
29	clinic	Kamala Clinic	11.8723673, 78.0874195	30 km/h	Medium
30	hospital	Kamala Hospital	11.8722588, 78.0899189	30 km/h	Medium
31	hospital	GJB Hospital	11.8713998, 78.0980288	30 km/h	Medium
32	clinic	Pranav Clinic	11.8712846, 78.0998167	30 km/h	Medium
33	hospital	Bethel Hospital	11.8699472, 78.1388598	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
24	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
26	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium

Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45853, 77.85139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45906, 77.85132



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45636, 77.81678



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89606



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49153, 77.89615



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.48364, 77.88780



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49155, 77.88776



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49441, 77.88495



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49469, 77.88488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.55614, 77.89764



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58330, 77.90642



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64217, 77.95200



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64712, 77.95118



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67858, 77.96834



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70218, 77.97866



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71917, 78.01123



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72565, 78.01770



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73787, 78.03528



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73651, 78.03868



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74120, 78.04588



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.78045, 78.05792



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.87104, 78.08356



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87106, 78.08579



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87257, 78.08738



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86902, 78.12044

